- A specific host on the network is specified by network ID as 0s and with required host ID. This is used as a source address.

At the last level, the two headers are option and padding. Option field in the IP header is optional. This field is made of option-type field, followed by option length field and then option data bytes. Option type is the first byte of option header, and it has three fields: 1-bit copied flag, 2-bit class field and 5-bit number field.

When copied flag bit is 0, option field is not copied in the fragments of the IP datagram. But when it is set to 1, option field is copied in all fragments of the IP datagram. The different option classes are as under:

00 for Control
01 for Reserved
10 for Debugging
11 for Reserved

where the numbers are used as:

0 for end of option list followed by 0 bytes

1 for no operation followed by 0 byte

2 for security followed by 10 bytes

3 for loose source and record routing (LSRR) followed by variable bytes (allowable maximum length is of 255 bytes in total)

4 for Internet timestamp followed by variable bytes, maximum of 255 bytes in total

7 record route (RR) followed by variable bytes, maximum of 255 bytes in total

8 for stream identification followed by 3 bytes

9 for strict source and record routing (SSRR) followed by variable bytes, maximum of 255 bytes in total

End-of-option list has type 0 as its option type; byte has a value of 0 (00000000). It belongs to class 00. This option field is used as the end of all options but not the end of each option. However, it does not necessarily coincide with the end of all IP headers as per the IP header length field. This is because padding is needed to make IP headers a multiple of 32-bit words. End-of-option field may be copied, introduced or deleted on fragmentation.

No operation option field is used between options for the purpose of alignment for 32-bit words. This field has a type that is equal to 1 (00000001) and belongs to control class 00. It may also be copied, introduced or deleted on fragmentation.

Security option field belongs to control class. The field has type 130 as its option type field and carries a value of 130 (10000010). The length field carries a value of 11 (00001011), as it has total 11 bytes.

Other option fields are SSS…SSS, CCC…CCC, HHH…HHH and TCC…TCC.

SSS is a 2-byte security field that specifies different levels of security. Examples are 00000000 00000000 for unclassified with code as 00110101 11100010 (reserved), 11110001 00110101 for confidential with code as 01001101 01111000 (reserved), 10101111 00010011 for restricted with code as 00010011 01011110 (reserved), 01101011 11000101 for top secret with code as 11000100 11010110 (reserved) and so on.

CCC field is also a 2-byte field. It is called compartment field. The field contains all 0s when data is not compartmented. Other codes are with Defence Intelligence Agency (DIA), USA.

HHH is a 2-byte field known as handling restrictions field. Code of the field is available from DIA.

TCC field is known as transmission control code, and is 3-byte long. It is used to segregate data and define control. It is also DIA-controlled. The field must be copied on fragmentation.

Loose source and record routing (LSRR) is the options field that allows source of an IP datagram to define routing information to be used by routers/gateways to forward the datagram to its destination. This is also used to record the routing information. Thus, the options field provides alternative routes. Routing is specified by recording the IP addresses of routers and gateways in the data area of the options field. Routers and gateways may use the information provided in the route data to route the datagram, and may not use their own routing tables for the same.

The Internet time stamp options field has headers like copied flag as 0, debugging class (10) with number 4 (00100). The option field is therefore type 68, as type header has a value of 68 (01000100). One byte of length header specifies the number of bytes in the options field, with a maximum value of 40. The 1-byte pointer header indicates the byte at which the time stamp is to be recorded. Minimum value of the pointer is 5, as the first time stamp record must start 5 bytes after the pointer (1 byte for overflow + flags, 4 bytes for Internet address).

The 4-bit overflow header is used to hold the number of IP addresses (may be routers, intermediate nodes, gateways, etc) that cannot record the timestamp for want of space. The 4-bit lags are used to define different modes of recording time stamps. The time stamp is of 4 bytes or 32-bit words that record value for the number of milliseconds since midnight under universal time frame. If non-standard time record is made, any time can be recorded, but in that case highest order bit of the time stamp is set to 1. The Internet address refers to the IP address of the source that started the option. END

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This article was originally published in the May 2019 issue of Electronics For You. The article is dedicated to Bob Metcalfe, who developed the Ethernet in 1973.